CDS 101 -Final Project Report

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1. Problem Definition

- Our project investigates whether weather conditions are related to equine incidents at New York race tracks. The main question we are studying is: How significant is the relationship between equine incidents and weather conditions?
- This topic is important because equine deaths and injuries are serious safety concerns in horse racing. By studying incident patterns over time and comparing them with weather conditions, we may be able to identify whether certain weather factors are associated with higher or lower incident counts.
- The objective of this analysis is to explore trends in equine incidents from 2009 to 2025, compare incidents across different years, tracks, incident types, seasons, and weather categories, and determine whether weather appears to have a noticeable relationship with incident patterns.
- A key assumption is that the incident reports are accurate and consistently recorded. Another assumption is that the weather information connected to each incident is reliable enough to be grouped into categories such as clear, rainy, cloudy, snowy, or unknown.

2. Data Acquisition & Description

- The name of the dataset is Equine Death and Breakdown from the data.gov website which is an official website of the United States government.
- Obtained through a download link from data.gov website.
- There are a lot of variables / features such as Incident Type, Horse, Jockey Driver, Track, and so forth but we will be focusing primarily on two variables, Death or Injury and Weather Conditions. Death or Injury variable shows if a horse died in an incident or was just injured. Weather Conditions variable shows the weather conditions at the time of the incident. We will also be using "Track", "Year", and "Incident Date" columns to help supplement our findings.
- The dataset has 5,383 rows and 13 columns.

3. Data Cleaning & Preprocessing

- This dataset was extremely messy and we had to do a lot of cleaning to make it usable. We got rid of any rows that had NA values in the Weather Condition and Death and Injury variables. The incident date variable had to be changed to a date type. The column names were fixed to make it easier to use and read. Originally the dataset's weather condition variable had 1000+ different unique entries but we wanted them to be categorized in just a few simple ones. The weather_clean variable that was created that holds the weather_conditions values without all the unnecessary information and spaces. This is then piped into weather_group which categorizes the specific weather condition happening at the time of the incident such as "Rain", "Snow", or "Clear" and so forth. The death_or_injury_clean variable cleans up the 10 unique values and categorizes them into 2 simple ones "death" or "injury." After creating these new variables we filtered them to only include the values that were relevant to our research question. A season variable was also created for EDA. Finally, the Weather Condition variable was cleaned so that way The Death and Injury variable had to be encoded to binary 0/1 variable since its a categorical variable and we need a numerical values to make a linear regression model.

```
#Fix column names
horses_clean <- horses %>%
  rename(year = Year, incident_date = `Incident Date`, track = Track, inv_location = `Inv Locat

#Fix dates
horses_clean$incident_date <- mdy(horses_clean$incident_date)

#Get rid of NA entries
horses_clean <- horses_clean %>%
  filter(!is.na(weather_conditions), !is.na(death_or_injury))

#Fix weather_condition column
#USED CHATGPT FOR THIS CODE!!!!!!!!!

horses_clean <- horses_clean %>%
  mutate(
    weather_clean = weather_conditions %>%
      str_to_lower() %>%
      str_trim() %>%
      str_replace_all("[0-9]+", "") %>%        # remove numbers
      str_replace_all("[^a-z ]", "") %>%       # remove symbols
      str_replace_all("couldy|cloudyy|clody", "cloudy") %>%
      str_replace_all("raine", "rain") %>%
      str_squish()
  ) %>%

  mutate(
    weather_group = case_when(
```

```r
      # Rain
      str_detect(weather_clean, "rain|shower|drizzle") ~ "Rain",

      # Snow
      str_detect(weather_clean, "snow|flurr|sleet") ~ "Snow",

      # Storms
      str_detect(weather_clean, "thunder|storm") ~ "Storm",

      # Cloudy / Overcast
      str_detect(weather_clean, "cloud|coudy|cloudy") ~ "Cloudy",

      # Clear / Sunny
      str_detect(weather_clean, "clear|sun|fair|dry") ~ "Clear",

      # Other weather conditions
      str_detect(weather_clean, "windy|fog|mist|hazy|wind") ~ "Other Weather",

      # Missing / junk
      weather_clean %in% c("", "na", "n a", "unc") | is.na(weather_clean) ~ "Unknown",

      TRUE ~ "Other"
    )
  )

#Clean death_or_injury column
horses_clean <- horses_clean %>%
  mutate(
    death_or_injury_clean = death_or_injury %>%
      str_to_lower() %>%
      str_trim(),

    death_or_injury_clean = case_when(
      str_detect(death_or_injury_clean, "death|dead|euthanasia") ~ "Death",
      str_detect(death_or_injury_clean, "lame|injury|lameness") ~ "Injury",
      TRUE ~ "Other"
    )
  )

horses_clean <- horses_clean %>%
  filter(death_or_injury_clean != "Other") %>%
  select(year, incident_date, incident_type, track, incident_description, weather_group, death_
  filter(weather_group != "Other" & weather_group != "Unknown")

#Create season variable (Code reference linked below)
horses_clean <- horses_clean %>%
  mutate(
```

```r
    season = case_when(
    month(incident_date) %in% 9:11 ~ "Fall",
    month(incident_date) %in% c(12,1,2) ~ "Winter",
    month(incident_date) %in% 3:5 ~ "Spring",
    TRUE ~ "Summer"
  ))
```

4. Exploratory Data Analysis (EDA)

This section maps directly to the **EDA** part of the rubric:

- Summary statistics
- Visualizations
- Interpretation connected to the research question

```r
summary(horses_clean)
```

```
##       year      incident_date        incident_type         track
##  Min.   :2009   Min.   :2009-04-28   Length:1702        Length:1702
##  1st Qu.:2012   1st Qu.:2012-01-20   Class :character   Class :character
##  Median :2016   Median :2016-10-01   Mode  :character   Mode  :character
##  Mean   :2017   Mean   :2017-03-27
##  3rd Qu.:2022   3rd Qu.:2022-09-15
##  Max.   :2026   Max.   :2026-04-12
##  incident_description weather_group      death_or_injury_clean
##  Length:1702          Length:1702        Length:1702
##  Class :character     Class :character   Class :character
##  Mode  :character     Mode  :character   Mode  :character
##
##
##
##     season
##  Length:1702
##  Class :character
##  Mode  :character
##
##
##
```

Incident Type

This section looks at the different types of racing incidents in the dataset. The goal is to see which incident types happen most often and whether those incident types changed over time.

```r
library(readr)

Equine_Death_and_Breakdown <- read_csv("data/Equine_Death_and_Breakdown.csv")
```

```
## Rows: 5381 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (12): Incident Date, Incident Type, Track, Inv Location, Racing Type Des...
## dbl  (1): Year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

#Number of incidents by incident type

```r
incident_summary <- as.data.frame(
  sort(table(Equine_Death_and_Breakdown[["Incident Type"]]), decreasing = TRUE)
)

names(incident_summary) <- c("Incident Type", "n")

incident_summary
```

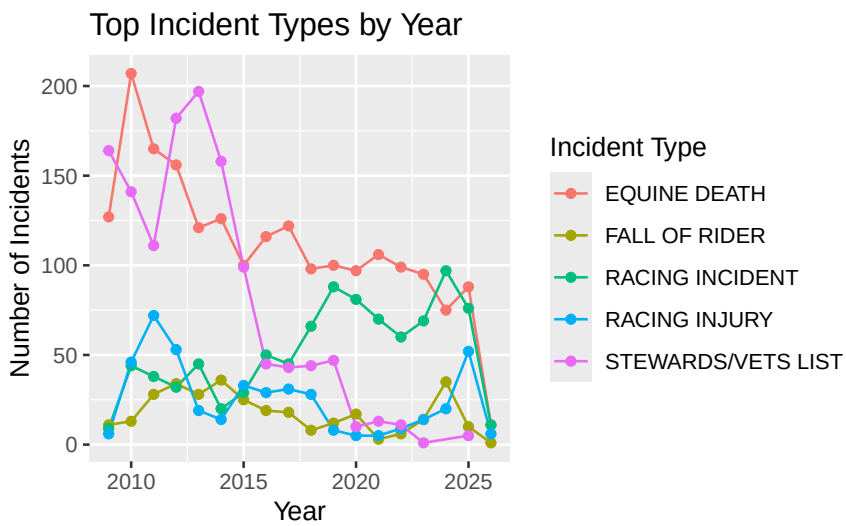
Incident Type	n
EQUINE DEATH	2009
STEWARDS/VETS LIST	1271
RACING INCIDENT	930
RACING INJURY	450
FALL OF RIDER	318
FALL OF HORSE	122
ACCIDENT - ON TRACK	89
TRAINING INCIDENT	68
ACCIDENT - TAGGED SULKY	57
ACCIDENT - IN STARTING GATE	22
ACCIDENT - DRIVER/JOCKEY	12
EQUINE DEATH - INFECTIOUS DISEASE	11
DRIVER/JOCKEY INJURED	9
ON-TRACK ACCIDENT	9
BREAKDOWN	4

*This graph connects to our research question by showing which types of equine incidents happen most often in the data set. Since our project is studying the relationship between weather related variables and equine deaths or breakdowns, we first need to understand what kinds of incidents are being recorded. The most common incident types are the main categories we can later compare

with weather conditions, temperature, precipitation, or season. This graph does not show a weather relationship by itself, but it gives important background for the weather analysis.

#Incident type by year

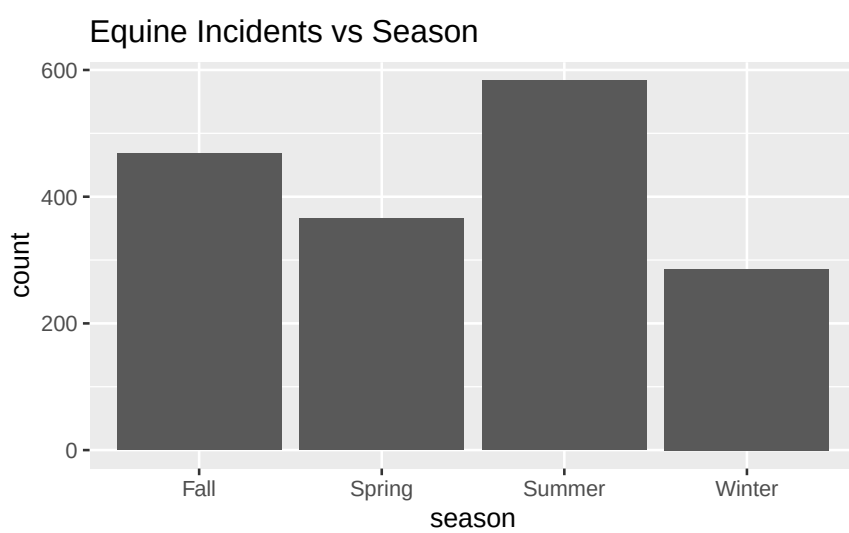
```
library(ggplot2)

top_incident_types <- incident_summary$`Incident Type`[1:5]

incident_year_data <- Equine_Death_and_Breakdown[
  Equine_Death_and_Breakdown[["Incident Type"]] %in% top_incident_types &
    !is.na(Equine_Death_and_Breakdown[["Year"]]),
]

incident_year_summary <- as.data.frame(
  table(
    Year = incident_year_data[["Year"]],
    Incident_Type = incident_year_data[["Incident Type"]]
  )
)

names(incident_year_summary) <- c("Year", "Incident Type", "n")

incident_year_summary <- incident_year_summary[incident_year_summary$n > 0, ]

incident_year_summary$Year <- as.numeric(as.character(incident_year_summary$Year))

ggplot(incident_year_summary,
       aes(x = Year, y = n, color = `Incident Type`, group = `Incident Type`)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Top Incident Types by Year",
    x = "Year",
    y = "Number of Incidents",
    color = "Incident Type"
  )
```



*This graph helps connect to our question by showing how incident types changed over time. If certain years had higher numbers of specific incident categories, those changes could later be compared with weather patterns from those same years. This helps us look for possible trends before testing whether weather-related variables are connected to equine deaths or breakdowns. The graph does not prove that weather caused the incidents, but it helps identify time-based patterns that may be useful for the rest of the project.

ii.

```
horses_clean %>%
  ggplot(aes(x = season)) +
  geom_bar() +
  labs(
    title = "Equine Incidents vs Season"
  )
```



- We assume that Winter has the most inclement weather out of all the seasons and yet we see that Winter has the least amount of Equine deaths and injuries which could indicate a weak correlation between season and Equine incidences if we take our assumption as the truth.

iii.

```
#Weather vs Incidents Graph
horses_clean %>%
ggplot(aes(x = death_or_injury_clean, fill = death_or_injury_clean)) +
  geom_bar() +
  facet_wrap(~ weather_group) +
  labs(
    x = "Death or Injury",
    title = "Equine Incidents vs Weather Condition")
```

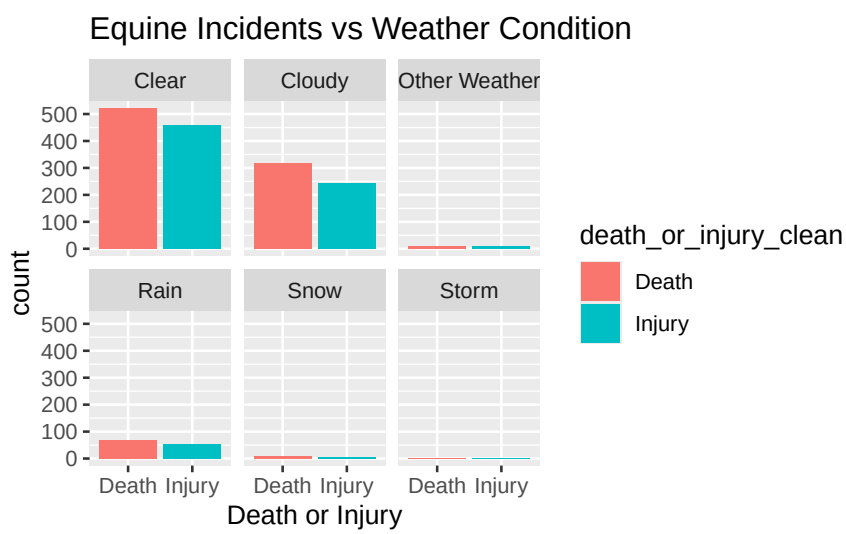


Figure 1: Example histogram of a numeric variable (EDA rubric).

- There does not appear to be a clear correlation between weather condition and Equine death or injury.

5. Visualization Quality and Storytelling

Use this section to satisfy the **Visualization Quality** rubric criterion:

- Explain why your plot types are appropriate.

- Comment on labels, legends, colors, and overall readability.

- Mention any steps you took to make plots accessible and interpretable.
- For the incident type section, we used a bar graph first because we are looking at incident type which is categorical.This graph makes sense because it lets us compare how many incidents happened in each incident type, and we flipped the graph so the incident type names are easier to read instead of crowded on the x-axis. For the second graph, we used a line graph because we are looking at incident type by year, which helps show how the incident categories changed over time instead of only showing the total number of incidents.
- For graphs ii. and iii. we used bar graphs because we were comparing two categorical variables to each other.

6. Modeling Approach

Explain how you framed the problem and which models you chose:

- Type of task (regression, classification, etc.).
- Baseline model or heuristic, if used.
- Main model(s) chosen and why they are appropriate.

7. Model Implementation & Evaluation

This corresponds to the **Model Implementation & Evaluation** rubric criterion.

Describe:

- Features used.
- Data splitting strategy (train/test or cross-validation).
- Metrics used (accuracy, F1, RMSE, etc.).
- Tables/plots summarizing performance.
- A short interpretation of each metric/plot.

```r
# Example structure:
# set.seed(123)
# train_index <- sample(seq_len(nrow(data_clean)), size = 0.8 * nrow(data_clean))
# train <- data_clean[train_index, ]
# test  <- data_clean[-train_index, ]
#
# model <- glm(target ~ ., data = train, family = binomial)
# preds <- predict(model, newdata = test, type = "response")
# # compute metrics...
```

8. Conclusions & Recommendations

Summarize the key takeaways:

- Answer your original research question(s) directly.
- Highlight the most important patterns or relationships you found.
- Discuss limitations (data size, bias, missing variables, etc.).
- Suggest possible extensions or future work.
- Possible

9. Code Quality & Reproducibility

Briefly document how someone else can reproduce your results:

- Which R scripts or Rmd files to run.
- Any required R packages.

-Explain how code is structured and include a link to your Git repository

10. References

List any references you used, such as:

- Dataset documentation.
- Research papers or articles.
- Tutorials or blog posts.

Dates to seasons column tutorial